

Section A — MCQs

1. **B — Group-2 (II-A)**
Third ionization energy is very high when an atom has 2 valence electrons.
 2. **C — Na & O**
Sodium is metal and oxygen is non-metal, so ionic bond forms.
 3. **A — $\text{H-Cl} > \text{H-F}$ (This is the False option, rest are True)**
 H-F bond is strongest due to small size and strong attraction.
 4. **C — 1.806×10^{24}**
 $67.2 \text{ dm}^3 \text{ He} = 3 \text{ moles}$.
 $\text{Particles} = 3 \times 6.02 \times 10^{23} = 1.806 \times 10^{24}$.
 5. **A — +3 kJ/mol**
 $\Delta H_{\text{solution}} = \text{lattice reverse} + \text{hydration}$
 $= +786 - 783 = +3 \text{ kJ/mol}$.
 6. **A — Increasing Temperature**
The reaction is already Exothermic, Since heat is a product, increasing temperature means adding more product (heat). So, equilibrium shifts backward to consume excess heat. Therefore, it does NOT shift forward.
 7. **C — CuSO_4**
 CuSO_4 solution is acidic due to hydrolysis of Cu^{2+} .
 8. **D — Group-14 (IV-A)**
The configuration $ns^2 np^2$ (valence shell) corresponds to Group 14 (Carbon family).
 9. **D — Unburned hydrocarbons**
PAN is formed in photochemical smog from hydrocarbons and nitrogen oxides.
 10. **D — NaI**
 I^- is strongest reducing halide ion and can reduce H_2SO_4 to H_2S .
 11. **B — Propagation**
Free radical attacks ethane and forms ethyl radical.
 12. **A — $\text{CH}_2=\text{CH}_2$**
Ethyl bromide with alcoholic NaOH gives ethene by elimination.
 13. **C — LiAlH_4**
Carboxylic acid is reduced to alcohol by LiAlH_4 .
 14. **B — Ethanal**
Ethanal gives both silver mirror test and iodoform test.
 15. **A — Ethanol and acetone**
2,4-DNPH identifies carbonyl compounds; acetone gives positive test, ethanol does not.
 16. **A — $t_{1/2} = 0.693/k$**
This is half-life formula for first-order reaction.
 17. **C — $\text{CH}_3\text{CH}_2\text{CHO}$**
Propanal is an aldehyde and gives Fehling's test.
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Section B — Short Questions

Q2(i) Electronic configuration of Cu^{2+} , As and Br^-

Cu = 29:

Neutral Cu = $[\text{Ar}] 3d^{10} 4s^1$

Cu^{2+} loses electrons first from 4s, then 3d:

$\text{Cu}^{2+} = [\text{Ar}] 3d^9$

As = 33:

As = $[\text{Ar}] 3d^{10} 4s^2 4p^3$

Br^- :

Br has 35 electrons, Br^- has 36 electrons:

$\text{Br}^- = [\text{Ar}] 3d^{10} 4s^2 4p^6$

OR: Catenation and its importance

Catenation is the ability of an element to form covalent bonds with itself.

Carbon shows strong catenation because C–C bond is strong and carbon has small size.

Importance: It allows carbon to form long chains, branched chains, rings and millions of organic compounds.

Q2(ii) Structure of PCl_5 and NH_4^+ by VSEPR

PCl_5 :

Phosphorus has 5 bond pairs and no lone pair.

Shape = **trigonal bipyramidal**

Bond angles = **120° and 90°**

NH_4^+ :

Nitrogen has 4 bond pairs and no lone pair.

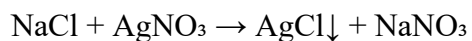
Shape = **tetrahedral**

Bond angle = **109.5°**

OR: Tests for Cl^- , Br^- and I^-

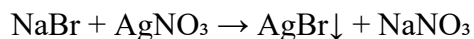
Add AgNO_3 solution.

Cl⁻:



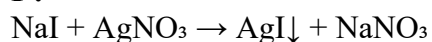
White precipitate, soluble in dilute NH_3 .

Br⁻:



Cream precipitate, partly soluble in conc. NH_3 .

I⁻:



Yellow precipitate, insoluble in NH_3 .

Q2(iii) Bond energy order

$\text{Cl}_2 > \text{F}_2$

F_2 has very small atoms, so lone pair-lone pair repulsion is very high. This weakens F–F bond. Therefore Cl–Cl bond is stronger.

$\text{H}_2 > \text{HBr}$

H–H bond is shorter and stronger than H–Br bond.

$\text{N}_2 > \text{O}_2$

N_2 has triple bond, while O_2 has double bond. Triple bond is stronger.

OR: Properties of water due to hydrogen bonding

1. Water has high boiling point.
2. Water has high surface tension.
3. Ice is less dense than water, so ice floats.

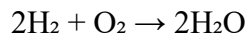
Reason: Water molecules are strongly connected by hydrogen bonds.

Q2(iv) Limiting and non-limiting reactant

Limiting reactant is the reactant that is completely consumed first and limits product formation.

Non-limiting reactant is present in excess and remains after the reaction.

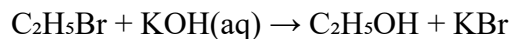
Example:



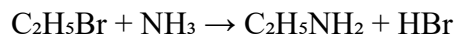
If H_2 is less, then H_2 is limiting and O_2 is excess.

OR: Ethyl bromide reactions

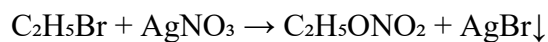
With aqueous KOH:



With NH_3 :



With AgNO_3 :



Q2(v) Crystalline and non-crystalline solids

Crystalline solids	Non-crystalline / amorphous solids
Regular arrangement	Irregular arrangement
Sharp melting point	No sharp melting point
Example: NaCl, diamond	Example: glass, rubber

OR: Shell, subshell and orbital

Shell: Main energy level, represented by n.

Example: K, L, M shells.

Subshell: Division of shell.

Example: s, p, d, f.

Orbital: Region where probability of finding electron is maximum.

Each orbital can hold maximum 2 electrons.

Q2(vi) Enthalpy of solution

For **HNO₃**, $\Delta H_{\text{solution}} = +25.7 \text{ kJ/mol}$.

Positive sign means heat is absorbed, so process is **endothermic**.

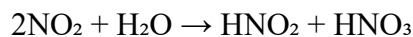
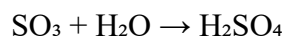
For **CaCl₂**, $\Delta H_{\text{solution}} = -83.0 \text{ kJ/mol}$.

Negative sign means heat is released, so process is **exothermic**.

OR: Acid rain formation

Acid rain is rain having pH less than 5.6.

It forms when oxides of sulphur and nitrogen dissolve in rainwater.



These acids fall with rain and damage buildings, plants and aquatic life.

Q2(vii) Measures to preserve water resources

1. Avoid throwing industrial waste into rivers.
2. Treat sewage water before discharge.
3. Use water carefully and avoid wastage.
4. Store rainwater where possible.

OR: Iodoform test for alcohols

Alcohols containing **CH₃–CH(OH)–** group give iodoform test.

Example: ethanol gives yellow precipitate of iodoform.

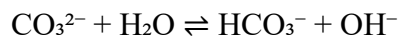


Yellow precipitate = **CHI₃**

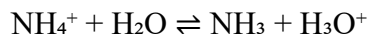
Q2(viii) Why Na₂CO₃ is basic and NH₄Cl is acidic?

Na₂CO₃ is salt of strong base NaOH and weak acid H₂CO₃.

CO₃²⁻ reacts with water to produce OH⁻ ions, so solution is basic.

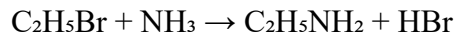


NH_4Cl is salt of weak base NH_3 and strong acid HCl .
 NH_4^+ reacts with water to produce H^+ ions, so solution is acidic.

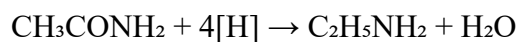


OR: Preparation of ethyl amine

From alkyl halide:



From amide:

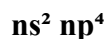


Reducing agent: LiAlH_4

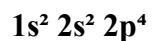
Q2(ix) Principal energy level, subshell and orbitals in Group VIA

Group VIA means group 16 elements.

Valence configuration:



Example oxygen:



For p^4 :

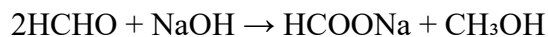
p orbitals are filled according to Hund's rule:



So group VIA elements have 6 valence electrons.

OR: Cannizzaro reaction of formaldehyde

Formaldehyde has no alpha hydrogen, so it gives Cannizzaro reaction.



One formaldehyde molecule is oxidized to formate ion, and the other is reduced to methanol.

Q2(x) Uses of sulphur compounds

1. Sulphuric acid is used in fertilizers and industry.
2. SO₂ is used as bleaching agent.
3. Sulphur compounds are used in medicines and fungicides.

OR: Markovnikov's rule

When an unsymmetrical reagent adds to an unsymmetrical alkene, hydrogen goes to the carbon already having more hydrogen atoms.

Example:



Major product = **2-bromopropane**

Reason: more stable secondary carbocation is formed.

Q2(xi) Volume of NH₃

Given molecules of NH₃ = 3.011×10^{20}

Avogadro number = 6.022×10^{23}

Moles:

$$\begin{aligned} &= 3.011 \times 10^{20} / 6.022 \times 10^{23} \\ &= 5 \times 10^{-4} \text{ mol} \end{aligned}$$

At STP:

$$1 \text{ mol gas} = 22.4 \text{ dm}^3$$

Volume:

$$\begin{aligned} &= 5 \times 10^{-4} \times 22.4 \\ &= 0.0112 \text{ dm}^3 \\ &= \mathbf{11.2 \text{ cm}^3} \end{aligned}$$

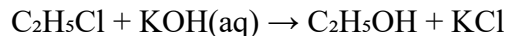
OR: Preparation of ethanol

From alkene:

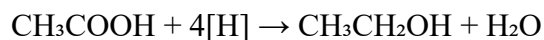


Condition: H_3PO_4 catalyst, heat and pressure.

From alkyl halide:



From carboxylic acid:

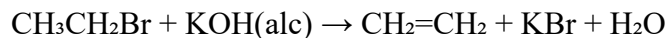


Reducing agent: LiAlH_4

Q2(xii) β -elimination reaction

In β -elimination, atoms are removed from adjacent carbon atoms and a double bond is formed.

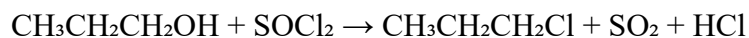
Example:



Reason: H is removed from β -carbon and Br is removed from α -carbon.

OR: 1-Propanol reactions

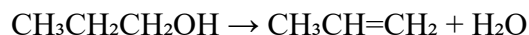
With SOCl_2 :



With PCl_5 :



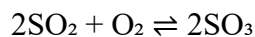
With conc. H_2SO_4 at 180°C :



Product = propene

Q2(xiii) Kc calculation

Reaction:



Given:

$$[\text{SO}_2] = 0.59 \text{ M}$$

$$[\text{O}_2] = 0.05 \text{ M}$$

$$[\text{SO}_3] = 0.259 \text{ M}$$

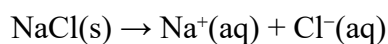
$$K_c = [\text{SO}_3]^2 / [\text{SO}_2]^2 [\text{O}_2]$$

$$K_c = (0.259)^2 / (0.59)^2 (0.05)$$

$$K_c = 0.067081 / 0.017405$$

$$K_c = \mathbf{3.85}$$

OR: Entropy change for NaCl dissolving



$$\Delta S = S \text{ products} - S \text{ reactants}$$

$$= [S^\circ \text{Na}^+ + S^\circ \text{Cl}^-] - S^\circ \text{NaCl}$$

$$= [21 + 56.5] - 72.1$$

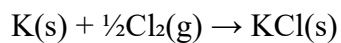
$$= 77.5 - 72.1$$

$$= \mathbf{+5.4 \text{ J mol}^{-1} \text{ K}^{-1}}$$

Positive value shows randomness increases on dissolving.

Q2(xiv) Born-Haber cycle for KCl

Formation reaction:



Steps:

1. Sublimation of K:
 $\text{K(s)} \rightarrow \text{K(g)}$
2. Ionization of K:
 $\text{K(g)} \rightarrow \text{K}^+(\text{g}) + \text{e}^-$
3. Atomization of chlorine:
 $\frac{1}{2}\text{Cl}_2(\text{g}) \rightarrow \text{Cl(g)}$
4. Electron affinity of chlorine:
 $\text{Cl(g)} + \text{e}^- \rightarrow \text{Cl}^-(\text{g})$
5. Lattice formation:
 $\text{K}^+(\text{g}) + \text{Cl}^-(\text{g}) \rightarrow \text{KCl(s)}$

Born-Haber cycle helps calculate lattice energy.

OR: Rate constant for N_2O_5 decomposition

Given:

N_2O_5 concentration decreases from 0.10 M to 0.05 M in 60 s.

This means one half-life:

$$t_{1/2} = 60 \text{ s}$$

For first order reaction:

$$t_{1/2} = 0.693 / k$$

$$k = 0.693 / 60$$

$$k = \mathbf{0.01155 \text{ s}^{-1}}$$

Section C — Long Questions

Q3: Mass spectrometry and average atomic mass

Mass spectrometry is a technique used to determine masses and relative abundance of isotopes.

Main steps

1. **Vaporization:** sample is converted into vapours.
2. **Ionization:** atoms are converted into positive ions.
3. **Acceleration:** ions are accelerated by electric field.
4. **Deflection:** ions are deflected according to mass/charge ratio.
5. **Detection:** detector records ions.

Lighter ions deflect more, heavier ions deflect less.

Average atomic mass

Average atomic mass is calculated using isotopic masses and their percentage abundance.

Formula:

$$\text{Average atomic mass} = \frac{\Sigma(\text{isotopic mass} \times \text{percentage abundance})}{100}$$

Example:

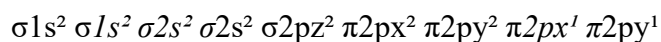
If chlorine has Cl-35 = 75% and Cl-37 = 25%:

$$\begin{aligned}\text{Average mass} &= (35 \times 75 + 37 \times 25) / 100 \\ &= 35.5 \text{ amu}\end{aligned}$$

OR: Bonding in O₂ by molecular orbital theory

O₂ has 16 electrons.

Molecular orbital configuration:



Bond order:

$$= \frac{1}{2}(\text{bonding electrons} - \text{antibonding electrons})$$

Bonding electrons = 10

Antibonding electrons = 6

$$\text{Bond order} = \frac{1}{2}(10 - 6) = 2$$

So O₂ has double bond.

Why MOT is better than VBT?

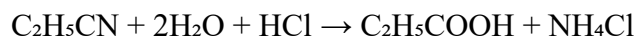
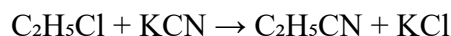
Valence bond theory predicts O₂ should have all paired electrons, but experimentally O₂ is paramagnetic.

MOT correctly explains that O₂ has two unpaired electrons in antibonding orbitals.

Therefore, MOT is superior for explaining magnetic behavior.

Q4: Preparation of carboxylic acids

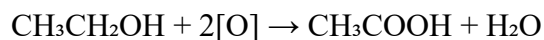
a. From ethyl chloride



Product = **propanoic acid**

Reason: KCN increases carbon chain by one carbon.

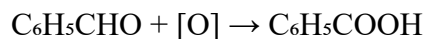
b. From ethanol



Oxidizing agent: acidified K₂Cr₂O₇ or KMnO₄

Product = **ethanoic acid**

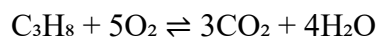
c. From benzaldehyde



Product = **benzoic acid**

OR: Effect of changes on equilibrium

i. Increasing pressure

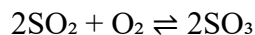


Reactant side has 6 gas moles, product side has 7 gas moles.

Increasing pressure shifts equilibrium toward fewer moles.

So equilibrium shifts **left**.

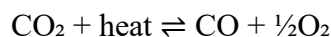
ii. Removing SO_3



SO_3 is product. Removing product shifts equilibrium forward.

So equilibrium shifts **right**.

iii. Decreasing temperature



Forward reaction is endothermic because heat is reactant.

Decreasing temperature shifts reaction backward to produce heat.

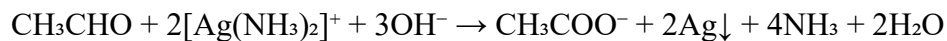
So equilibrium shifts **left**.

Q5: Identification of functional groups

a. CH_3CHO

Functional group: **aldehyde**

Test: Tollens' test

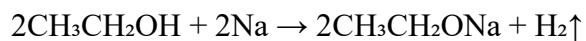


Observation: silver mirror.

b. CH₃CH₂OH

Functional group: **alcohol**

Test: sodium metal test

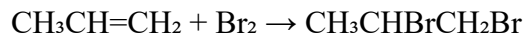


Observation: hydrogen gas bubbles.

c. CH₃CH=CH₂

Functional group: **alkene**

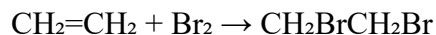
Test: bromine water



Observation: reddish-brown bromine water is decolorized.

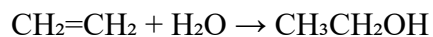
OR: Ethane and ethene**Test to distinguish ethane and ethene**

Ethene decolorizes bromine water, but ethane does not.

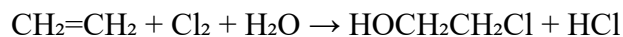


Reason: ethene has C=C double bond and undergoes addition reaction.

Ethane is saturated, so it does not react under normal conditions.

Reactions of ethene**With H₂SO₄/H₂O:**

Product = ethanol

With Cl₂/H₂O:

Product = ethylene chlorohydrin

Q6: Solubility product of CaF_2



Solubility product:

$$K_{\text{sp}} = [\text{Ca}^{2+}][\text{F}^-]^2$$

If molar solubility of CaF_2 is s :

$$[\text{Ca}^{2+}] = s$$

$$[\text{F}^-] = 2s$$

$$K_{\text{sp}} = s(2s)^2$$

$$K_{\text{sp}} = 4s^3$$

Condition for precipitation

Ionic product:

$$Q_{\text{sp}} = [\text{Ca}^{2+}][\text{F}^-]^2$$

If $Q_{\text{sp}} > K_{\text{sp}}$, precipitation occurs.

If $Q_{\text{sp}} < K_{\text{sp}}$, no precipitation occurs.

If $Q_{\text{sp}} = K_{\text{sp}}$, solution is saturated.

So CaF_2 precipitates when product of Ca^{2+} and F^- ions exceeds K_{sp} .

OR: Collision theory and reaction conditions

According to collision theory, a chemical reaction occurs only when reacting particles collide with:

1. Proper orientation
2. Energy equal to or greater than activation energy

Not every collision gives product. Only effective collisions produce reaction.

Activation energy

Activation energy is the minimum energy required for reactants to form activated complex.

Activated complex

It is the unstable high-energy intermediate formed during reaction.

Reaction path:

Reactants $\rightarrow$ Activated complex $\rightarrow$ Products

Energy change

If products have lower energy than reactants, reaction is exothermic.

If products have higher energy than reactants, reaction is endothermic.